

to large volumes of failed transactions for which they would not receive payment.

The gas price is determined by the market and effectively creates an auction for inclusion in the next Ethereum block. Higher gas fees signal higher demand and therefore generally receive higher priority for inclusion.

A technical aside about transactions is that they are posted to a *memory pool*, or *mempool*, before they are added to a block. Miners monitor these posted transactions, add them to their own mempool, and share the transaction with other miners to be included in the next available block. If the gas price offered by the transaction is uncompetitive relative to other transactions in the mempool, the transaction is deferred to a future block.

Any actor can see transactions in the mempool by running or communicating with mining nodes. This visibility can even allow for advanced front-running and other competitive techniques that aid the miner in profiting from trading activity. In contrast to traditional centralized markets, this front-running is legal given that all information is public. If miners see a transaction in the mempool, they could profit from it either by executing it themselves or front-running it and is incentivized to do so if they are lucky enough to win the block. Any occurrence of direct execution is known as *miner extractable value* (MEV), which is a drawback to the proof-of-work model. Certain strategies, such as obfuscating transactions, can mitigate MEV, thus hiding from miners how they might profit from the transactions.